Predicting Traffic Accident Severity in Bahrain: A Comparative Study of Classification Methods

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Abstract

Traffic accidents represent a critical public safety challenge in Bahrain, resulting in significant casualties and economic losses annually. This study investigates the application of supervised learning classification techniques to predict accident severity levels (Fatal, Serious, and Slight) using historical traffic accident data from Bahrain. We implemented and compared four classical classification methods: Logistic Regression, Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and K-Nearest Neighbors (KNN). The dataset comprised 35,999 accident records with features including temporal information, geographic coordinates, weather conditions, road characteristics, and casualty information. All four models achieved identical accuracy of 79.95%, matching the majority class baseline. Detailed analysis revealed that severe class imbalance (79.95% Slight, 15.08% Serious, 4.97% Fatal) caused models to default to majority class prediction, resulting in zero sensitivity for Fatal and Serious accidents. This study demonstrates a critical limitation of standard classification methods when applied to imbalanced datasets: high overall accuracy can mask complete failure to detect minority classes. While the implemented models are unsuitable for deployment, this work provides valuable insights into the challenges of imbalanced classification and establishes concrete recommendations including SMOTE oversampling, class weighting, ensemble methods, and alternative evaluation metrics focused on minority class performance.

I. Introduction

Road traffic accidents constitute a major public health concern globally, causing approximately 1.35 million deaths annually according to the World Health Organization. In Bahrain, despite significant infrastructure improvements and traffic management initiatives, accidents remain a persistent challenge affecting public safety and economic productivity. The severity of these accidents varies considerably, ranging from minor collisions resulting in slight injuries to catastrophic events causing fatalities.

Accurate prediction of accident severity is crucial for multiple stakeholders. Emergency medical services can optimize resource allocation and response strategies when severity can be anticipated. Traffic management authorities can identify high-risk conditions and implement targeted preventive measures. Policy makers can develop evidence-based regulations to enhance road safety. Traditional approaches to accident analysis have been largely reactive and descriptive. However, advances in machine learning and statistical learning provide opportunities to develop predictive models that can classify accident severity based on contributing factors.

Previous research has demonstrated the effectiveness of various classification techniques in traffic accident analysis. Kumar and Toshniwal (2016) applied support vector machines and decision trees to predict accident severity in India, achieving accuracies above 80%. Wahab and Jiang (2019) used random forests to analyze factors contributing to severe accidents, identifying speed limit and number of vehicles as critical predictors. However, many studies have not adequately addressed the challenge of class imbalance, which is inherent in accident data where severe accidents are naturally less frequent than minor ones. This imbalance poses significant methodological challenges that can lead to misleadingly high accuracy metrics while failing to detect the most critical cases.

Class imbalance occurs when the distribution of classes in a dataset is highly uneven, with one or more majority classes having significantly more instances than minority classes. In supervised learning, standard classification algorithms aim to minimize overall error during training. When faced with severe imbalance, models often learn to predict the majority class for all or most instances, as this strategy maximizes overall accuracy with minimal learning. The fundamental issue is that accuracy treats all errors equally, whereas in domains like traffic safety, failing to predict a fatal accident has far greater consequences than misclassifying a slight one.

This study adopts a systematic approach to classify traffic accident severity in Bahrain using four established statistical learning methods: Multinomial Logistic Regression, Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and K-Nearest Neighbors (KNN). These methods were selected based on course requirements and their theoretical foundations in classification theory. Our analysis examines not only predictive accuracy but also the practical challenges of implementing these methods on imbalanced real-world data, providing insights into why standard approaches fail and how they can be improved.

The rest of this paper is structured as follows. Section 2 describes the study data and provides a comprehensive statistical summary, with particular attention to class distribution. Section 3 details the implemented classification methods and their theoretical foundations. Section 4 presents the experimental results and performance comparison, while Section 5 discusses the implications of our findings, the root causes of model failure, and concrete strategies for addressing class imbalance. Finally, Section 6 concludes the study and highlights directions for improvement.

II. Exploratory Data Analysis

A. Data Collection

The dataset used in this study consists of traffic accident records from Bahrain spanning multiple years. The original data was obtained from official traffic authorities and contains comprehensive information about accident characteristics, environmental conditions, and outcomes. The dataset includes 35,999 accident records, each representing a unique traffic incident with associated attributes.

The target variable is accident severity, classified into three categories: Fatal (accidents resulting in at least one death), Serious (accidents causing severe injuries requiring hospitalization), and Slight (accidents resulting in minor injuries or property damage only). This three-class classification scheme aligns with international standards for accident severity reporting and provides actionable categorization for emergency response planning.

B. Data Processing

The raw dataset underwent several preprocessing steps to ensure data quality and suitability for statistical modeling. First, we performed data cleaning to handle missing values and inconsistencies. The dataset exhibited minimal missing data, with complete records for all critical variables. Categorical variables such as Weather, Road Type, and Light Conditions were converted to factor variables for proper handling in R.

Feature engineering was conducted to enhance predictive power. We created a binary indicator variable (Is_Night) derived from the Hour variable to capture day-night patterns more explicitly. This feature distinguishes between accidents occurring during daylight hours (6 AM to 6 PM) and nighttime hours (6 PM to 6 AM), which research suggests have different risk profiles.

Numeric variables including Year, Speed Limit, Vehicles Involved, Casualties, Hour, Latitude, and Longitude were retained in their original scales for Logistic Regression, LDA, and QDA. For the KNN classifier, which is sensitive to feature scales, we applied standardization (centering and scaling) to ensure all numeric features contribute equally to distance calculations.

The dataset was randomly partitioned into training and test sets using a 70-30 split, stratified by the target variable to maintain class proportions. This resulted in 25,200 training instances and 10,799 test instances. The stratified split is particularly important for imbalanced data, as it ensures that minority classes are adequately represented in both sets. The training set was used for model fitting and parameter selection, while the test set was reserved exclusively for final performance evaluation to provide unbiased accuracy estimates.

C. Data Summary

Table 1 presents the distribution of accident severity across the dataset. The data exhibits severe class imbalance, with Slight accidents comprising 79.95% of all cases, Serious accidents accounting for 15.08%, and Fatal accidents representing only 4.97%. This represents approximately a 16:1 ratio between the most common class (Slight) and the rarest class (Fatal).

Table 1. DISTRIBUTION OF ACCIDENT SEVERITY

Severity	Count	Percentage	Training Set	Test Set
slight	28,782	79.95%	20,148	8,634
serious	5,428	15.08%	3,800	1,628
fatal	1,789	4.97%	1,252	537
Total	**35,999**	**100%**	**25,200**	**10,799**

This severe imbalance poses significant challenges for classification algorithms, which tend to be biased toward the majority class. Standard classification methods optimize for overall accuracy, and when 80% of the data belongs to one class, a naive model that always predicts "Slight" would achieve 80% accuracy without learning any patterns. This "No Information Rate" becomes the baseline against which our models must be evaluated.

Figure 1 illustrates the temporal distribution of accidents by hour of day. The data reveals distinct patterns with peak accident frequencies during morning rush hours (7-9 AM) and evening rush hours (5-7 PM), corresponding to high traffic volumes during commuting periods. Nighttime hours (midnight to 5 AM) show lower accident frequencies but potentially higher severity rates, warranting the inclusion of time-based features in predictive models.

The geographic distribution of accidents shows concentration in urban areas, particularly around major highways and commercial districts. Speed limit analysis reveals that most accidents occur in zones with speed limits between 80-100 km/h, suggesting highways and major roads as high-risk locations. Weather conditions were predominantly clear (85% of accidents), with smaller proportions occurring during rain, fog, or other adverse conditions.

Correlation analysis among numeric predictors indicates moderate positive correlation between Vehicles Involved and Casualties ($r = 0.42$), as expected since multi-vehicle collisions tend to result in more casualties. Speed Limit shows weak negative correlation with accident frequency in urban areas, likely reflecting the fact that urban areas with lower speed limits have higher accident frequencies but potentially lower severity compared to high-speed rural roads.

D. Understanding Class Imbalance

A critical observation from the exploratory analysis is the relationship between class imbalance and model evaluation metrics. Class imbalance is a common issue in classification problems where the distribution of classes is highly uneven. In our dataset, the 16:1 ratio between Slight and Fatal accidents creates several specific challenges:

1. Model Bias Toward Majority Class: In supervised learning, models minimize overall error during training. With 80% of training examples labeled "Slight," the model can achieve high accuracy (approximately 80%) by simply predicting "Slight" for all instances, regardless of actual

patterns in the data. This is mathematically the optimal strategy when the objective is to maximize accuracy and all misclassification costs are equal.

2. Insufficient Minority Class Examples: While 1,789 Fatal accidents may seem like a reasonable sample size in absolute terms, it represents only 5% of the data. Models have limited examples from which to learn the distinctive patterns of fatal accidents. Moreover, if there is heterogeneity within the Fatal class (different types of fatal accidents with different characteristics), the effective sample size for learning each pattern is even smaller.

3. Misleading Accuracy Metrics: Traditional accuracy metrics exacerbate the problem by not accounting for class rarity. A model that achieves 79.95% accuracy by always predicting "Slight" appears successful but provides zero predictive value. The model has not learned anything useful—it has merely memorized the majority class label.

4. Failure to Detect Critical Events: From a practical perspective, fatal and serious accidents are precisely the cases where prediction would be most valuable. A model that cannot detect any fatal accidents, regardless of its overall accuracy, is useless for the intended application of improving emergency response and reducing casualties. This insight motivated our decision to examine confusion matrices, class-specific sensitivity measures, and the Kappa statistic in addition to overall accuracy. The Kappa statistic measures agreement between predicted and actual classifications while adjusting for agreement expected by chance. A Kappa value of 0 indicates no agreement beyond chance, while values above 0.6 indicate substantial agreement. As we will see in the results section, all our models achieved Kappa values near zero, confirming that they provide no value beyond baseline guessing.

III. Methodology

This study adopted four classification models, each with distinct theoretical foundations and assumptions. The following subsections briefly describe each method and justify their implementation for this study.

A. Model 1: Multinomial Logistic Regression

Logistic regression extends linear regression to classification problems by modeling the log-odds of class membership as a linear function of predictor variables. For multi-class problems with K categories, multinomial logistic regression models the probability of each class relative to a baseline category.

The model assumes that the log-odds of belonging to class k versus the reference class is linear in the predictors:

$$\log(P(Y = k) / P(Y = \text{reference})) = \beta_{0k} + \beta_{1k}X_1 + \beta_{2k}X_2 + ... + \beta_{pk}X_p$$

where Y is the response variable (accident severity), X_1, X_2, ..., X_p are predictor variables, and β coefficients are estimated using maximum likelihood estimation. Predictions are made by calculating the probability for each class and assigning the observation to the class with highest probability.

We implemented multinomial logistic regression using the multinom function from the nnet package in R. The model included all available predictors: Year, Weather, Road Type, Light Conditions, Speed Limit, Vehicles Involved, Casualties, Hour, Latitude, Longitude, and Is_Night. Maximum iterations were set to 1,000 to ensure convergence.

Logistic regression is advantageous because it provides probabilistic predictions and interpretable coefficients indicating the effect of each predictor on class membership. It makes minimal distributional assumptions about predictors, unlike discriminant analysis methods. However, it assumes linear decision boundaries and may struggle with complex non-linear relationships. Importantly for our study, standard logistic regression does not account for class imbalance unless class weights are explicitly specified.

B. Model 2: Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis is a classical classification method that assumes predictors follow a multivariate normal distribution within each class, with all classes sharing a common covariance matrix. LDA finds linear combinations of predictors that best separate the classes.

The method estimates the following for each class k:

- Prior probability π_k (proportion of training observations in class k)
- Class-specific mean vector μ_k
- Common covariance matrix Σ

The discriminant function for class k is given by:

$$\delta_k(x) = x^T \Sigma^{-1}\mu_k - \tfrac{1}{2}\mu_k^T \Sigma^{-1}\mu_k + \log(\pi_k)$$

An observation is classified to the class with the largest discriminant function value. The decision boundaries between classes are linear, hence the name Linear Discriminant Analysis.

We implemented LDA using the lda function from the MASS package in R. The model utilized all predictor variables. LDA's assumption of common covariance matrices makes it computationally efficient and generally robust, particularly when the assumption holds approximately. However, when class covariances differ substantially, QDA may provide better performance. In the context of class imbalance, LDA uses the observed class proportions as prior probabilities, which means the model is inherently biased toward the majority class.

C. Model 3: Quadratic Discriminant Analysis (QDA)

Quadratic Discriminant Analysis relaxes the LDA assumption of equal covariance matrices, allowing each class to have its own covariance matrix Σ_k. This flexibility enables QDA to model more complex decision boundaries.

The discriminant function for QDA includes quadratic terms:

$$\delta_k(x) = -\tfrac{1}{2}\log|\Sigma_k| - \tfrac{1}{2}(x - \mu_k)^T \Sigma_k^{-1}(x - \mu_k) + \log(\pi_k)$$

The decision boundaries between classes become quadratic surfaces rather than hyperplanes. This increased flexibility comes at the cost of estimating more parameters (a separate covariance matrix for each class), which can lead to higher

variance in parameter estimates, especially with limited training data or high-dimensional feature spaces.

We implemented QDA using the qda function from the MASS package. QDA is theoretically superior to LDA when class covariances genuinely differ. However, the additional parameters increase the risk of overfitting, particularly for minority classes with fewer training examples. In our case with severe class imbalance, the Fatal class has only 1,252 training observations from which to estimate a full covariance matrix, which may lead to unstable estimates.

D. Model 4: K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a non-parametric classification method that makes no assumptions about the underlying data distribution. For a test observation, KNN identifies the K training observations closest to it in feature space and assigns the test observation to the most common class among these neighbors.

The classification rule is:
$P(Y = j \mid X = x_0) = (1/K) \Sigma_{i \in N_0} I(y_i = j)$
where N_0 is the neighborhood of x_0 containing the K nearest neighbors, and $I(\cdot)$ is the indicator function.
The choice of K is critical: small K values lead to flexible but high-variance predictions (sensitive to noise), while large K values produce smoother but potentially high-bias predictions. We used cross-validation on the training set to select the optimal K, testing odd values from 1 to 31 to avoid ties.

Since KNN relies on distance calculations, feature scaling is essential. We standardized all numeric predictors (Year, Speed Limit, Vehicles Involved, Casualties, Hour, Latitude, Longitude, and Is_Night) to have mean 0 and standard deviation 1. Categorical variables were excluded from the KNN model due to the difficulty of defining meaningful distances for nominal variables.

KNN's advantages include flexibility to capture complex non-linear patterns and simplicity (no training phase beyond storing data). Disadvantages include computational cost at prediction time, sensitivity to irrelevant features, and poor performance in high-dimensional spaces due to the "curse of dimensionality." In the context of class imbalance, KNN can be particularly problematic because neighborhoods in the feature space are likely to be dominated by majority class instances, leading to systematic bias in predictions.

E. Performance Measures

Model performance was evaluated using several metrics computed from confusion matrices on the test set:

1. Overall Accuracy: The proportion of correctly classified instances across all classes. While intuitive, accuracy can be misleading for imbalanced datasets.

2. Confusion Matrix: A table showing predicted versus actual classes, providing insight into which classes are confused with others.

3. Sensitivity (Recall): The proportion of actual class members correctly identified, computed separately for each class. For example, Fatal sensitivity = (Correctly predicted Fatal) / (Total actual Fatal). High sensitivity for Fatal and Serious classes is critical from a public safety perspective.

4. Specificity: The proportion of non-members correctly identified as such for each class.

5. Kappa Statistic: Measures agreement between predicted and actual classifications, adjusting for agreement expected by chance. Kappa = 0 indicates no agreement beyond chance, while Kappa = 1 indicates perfect agreement. Values above 0.6 are typically considered substantial agreement.

6. No Information Rate (NIR): The accuracy achieved by always predicting the majority class. Comparing model accuracy to NIR indicates whether the model provides value beyond naive prediction.

7. Balanced Accuracy: The average of sensitivity and specificity for each class, providing a measure that is less sensitive to class imbalance than overall accuracy.

These metrics collectively provide a comprehensive view of model performance, particularly important given the class imbalance in our data. As we will demonstrate in the results section, examining only overall accuracy would lead to incorrect conclusions about model effectiveness.

IV. Results and Discussion

This section reports and discusses the results of the four implemented classification models. We present overall performance metrics, detailed confusion matrices, and class-specific performance measures, with particular emphasis on understanding why all models failed to provide useful predictions.

A. Overall Model Comparison

Table 2 presents the overall accuracy of each model on the test set. Remarkably, all four models achieved nearly identical accuracy of 79.95%, with 95% confidence intervals of (0.7953, 0.8036).

TABLE 2. MODEL PERFORMANCE COMPARISON

Model	Accuracy	95%CI	Kappa	No info Rate
Logistic Regression	0.7995	(0.7953, 0.8036)	0.0000	0.7995
LDA	0.7995	(0.7953, 0.8036)	0.0000	0.7995
QDA	0.7995	(0.7953, 0.8036)	0.0000	0.7995
KNN	0.7995	(0.7953, 0.8036)	0.0000	0.7995

At first glance, these results might suggest successful classification with approximately 80% accuracy. However, critical examination reveals several alarming indicators:

First, the accuracy of 79.95% exactly matches the No Information Rate (the proportion of the majority class). This is not a coincidence—it strongly suggests that models are simply predicting the majority class for all or nearly all observations.

Second, the Kappa statistic is 0.0000 (or slightly negative for KNN), indicating zero agreement beyond what would be expected by chance alone. A Kappa of 0 signifies that the model performs no better than random guessing after accounting for class imbalance. For comparison, Kappa values above 0.6 indicate substantial agreement, and values above 0.8 indicate excellent agreement. Our models fall far short of any meaningful predictive performance.

Third, all models achieved identical performance despite their different theoretical foundations and assumptions. Logistic regression uses probabilistic linear models, LDA and QDA use discriminant analysis with different covariance assumptions, and KNN is a non-parametric distance-based method. The fact that such diverse approaches produce identical results is unusual and suggests a common underlying issue rather than genuine predictive performance.

Fourth, the p-value for testing whether accuracy exceeds the No Information Rate is approximately 0.50 for all models, indicating no statistical evidence that our models perform better than the baseline of always predicting the majority class.

B. Confusion Matrix Analysis

Table 3 presents the confusion matrix for Logistic Regression, which is representative of all models since they produced virtually identical predictions.

TABLE 3. CONFUSION MATRIX (ALL MODELS)

Actual	Predicted: Fatal	Predicted: Serious	Predicted: Slight
Fatal	0	0	1,789
Serious	0	0	5,428
Slight	0	0	28,782

This confusion matrix reveals the critical failure of all models: every single test observation was classified as "Slight," regardless of actual severity. The diagonal entries for Fatal and Serious classes are zero, meaning not a single instance of these critical classes was correctly identified. The models achieved perfect specificity for Fatal and Serious classes (100%—no false positives) but zero sensitivity (0%—no true positives). Conversely, the Slight class achieved 100% sensitivity (all Slight accidents were classified as Slight) but 0% specificity (all Fatal and Serious accidents were also misclassified as Slight).

C. Class-Specific Performance Metrics

Table 4 breaks down performance metrics by class, highlighting the severe deficiency in minority class prediction.

Table 4. CLASS-SPECIFIC PERFORMANCE METRICS

class	sensitivity	specificity	Pos Pred Value	Neg Pred Value	Prevalence	Balanced Accuracy
Fatal	0.000	1.0000	NaN	0.9503	0.0497	0.5000
Serious	0.000	1.0000	NaN	0.8492	0.1508	0.5000
Slight	1.0000	0.0000	0.7995	NaN	0.7995	0.5000

Zero sensitivity for Fatal and Serious classes means the models failed to identify any severe accidents whatsoever. From a practical standpoint, this represents complete failure for the most critical use case. Emergency services would receive no advance warning of severe accidents, defeating the primary purpose of predictive modeling in this domain. A system that cannot identify fatal or serious accidents has no practical value, regardless of its overall accuracy.

The Positive Predictive Value (precision) for Fatal and Serious classes is undefined (NaN) because the models never predicted these classes—resulting in division by zero in the precision calculation. This is another indicator of complete failure.

Balanced Accuracy, which averages sensitivity and specificity, is exactly 0.5 for all classes. This is equivalent to random guessing—a coin flip would perform equally well.

D. Root Cause Analysis: Why All Models Failed

The uniform poor performance across all models points to a common cause: severe class imbalance. Several interconnected factors contributed to this failure:

1. Objective Function Mismatch

Classification algorithms optimize for overall accuracy (or equivalently, minimize total misclassification errors). In imbalanced settings, this objective is dominated by the majority class. Consider the optimization from the model's perspective: by predicting "Slight" for all instances, the model achieves 79.95% accuracy immediately. To improve beyond this baseline, the model would need to correctly identify minority class instances at a rate that compensates for the false positives it creates. Given that minority classes constitute only 20% of the data, this is mathematically challenging.

Specifically, if the model attempts to identify Fatal accidents, each false positive (incorrectly predicting Fatal for a Slight accident) counts as one error, while each true positive (correctly predicting Fatal) reduces errors by one. However, because Fatal accidents are rare, the model encounters far more opportunities to make false positive errors than to make true positive gains. The result is that the accuracy-maximizing strategy becomes "always predict Slight."

2. Insufficient Minority Class Representation

Fatal accidents constitute only 4.97% of the training data (1,252 observations). While this is not trivially small in absolute terms, it may be insufficient for models to learn distinctive patterns, especially given potential heterogeneity

within the Fatal class. Fatal accidents likely occur under diverse circumstances—high-speed collisions, pedestrian accidents, multi-vehicle pileups, etc.—each with different characteristic patterns. With only ~1,250 examples distributed across these various scenarios, the effective sample size for learning any particular pattern is much smaller.

Moreover, many of the features that truly determine severity—such as impact speed, angle of collision, vehicle safety features, seatbelt use, driver age and condition—are not available in our dataset. Without sufficiently discriminative features, even sophisticated models struggle to separate classes.

3. No Class-Specific Handling
None of the implemented methods included mechanisms to address class imbalance. The implementations used default settings without:

- Class weighting: Penalizing misclassification of minority classes more heavily
- Synthetic oversampling: Creating artificial examples of minority classes (e.g., SMOTE)
- Undersampling: Reducing majority class size to balance distributions
- Cost-sensitive learning: Assigning different costs to different types of misclassification
- Threshold adjustment: Modifying decision thresholds to favor minority class prediction

These techniques are essential for imbalanced classification but were not applied in accordance with course requirements to implement standard methods.

4. Model-Specific Considerations

- Logistic Regression and Discriminant Analysis: These methods use class proportions from the training data, either explicitly as prior probabilities (LDA/QDA) or implicitly through the optimization process (Logistic Regression). With 80% of training data being Slight, the models are inherently biased toward this class.

- KNN: In imbalanced datasets, the K-nearest neighbors of any point are likely to be dominated by majority class instances simply due to their prevalence. Even if a test observation is actually Fatal, if it is in a region of feature space where Slight accidents are common, most of its neighbors will be Slight, leading to a Slight prediction.

5. The Misleading Nature of Accuracy
This study demonstrates an important lesson about evaluation metrics. Overall accuracy of 79.95% sounds impressive and might be reported as successful in less careful analyses. However, this accuracy is entirely attributable to correct classification of the majority class. The models provide zero information about Fatal and Serious accidents—the very cases where prediction would be most valuable.

This highlights why the Kappa statistic is crucial for imbalanced problems. By adjusting for chance agreement, Kappa reveals that our 79.95% accuracy is no better than random guessing given the class distribution. In domains where minority classes are critically important, accuracy alone is an inappropriate and potentially dangerous metric.

E. Comparative Analysis: Why Different Models Produced Identical Failures

An unexpected finding of this study is that four theoretically distinct classification methods—representing different statistical paradigms—all failed in exactly the same way. This section analyzes why each model failed and what this uniform failure reveals about the nature of the class imbalance problem.

Summary of Model-Specific Vulnerabilities to Class Imbalance:

• **Logistic Regression** - Probabilistic linear model using maximum likelihood estimation. Expected to provide flexible, interpretable predictions without distributional assumptions. Failed because likelihood maximization is dominated by majority class; with 80% Slight accidents in training data, parameters are pulled toward "always Slight" predictions to maximize overall likelihood. Imbalance sensitivity: **Very High** (directly uses class frequencies in optimization).

• **Linear Discriminant Analysis (LDA)** - Gaussian discriminant analysis with equal covariance matrices. Expected to provide efficient parameter estimation when normality assumptions hold. Failed because prior probabilities are set to training proportions (80% Slight), causing discriminant functions to be inherently biased toward majority class through the $\log(\pi)$ term. Imbalance sensitivity: **Very High** (explicitly uses empirical class priors).

• **Quadratic Discriminant Analysis (QDA)** - Gaussian discriminant with class-specific covariances. Expected to capture different variance structures across classes with more flexible decision boundaries than LDA. Failed because insufficient Fatal/Serious data (only 1,252 and 3,800 examples) leads to unstable covariance matrix estimation, while prior probabilities remain biased toward majority. Imbalance sensitivity: **Very High** (compounds LDA's prior problem with parameter instability).

• **K-Nearest Neighbors (KNN)** - Non-parametric distance-based method using majority voting. Expected to capture local patterns without distributional assumptions. Failed because neighborhoods are dominated by Slight accidents due to their prevalence in feature space; even true Fatal/Serious instances are surrounded by Slight neighbors, leading to Slight votes. Imbalance sensitivity: **Extremely High** (directly reflects class frequency in local regions).

Detailed Model-by-Model Analysis:

1. Logistic Regression: The Probability Trap
Multinomial logistic regression models the log-odds of each class relative to a baseline. The maximum likelihood estimation process adjusts coefficients to maximize the probability of observing the training data. With 20,148

Slight accidents versus 1,252 Fatal accidents in the training set (16:1 ratio), the likelihood function is dominated by correctly classifying Slight accidents.

Mathematically, the log-likelihood is:
$L = \Sigma_i \log P(y_i \mid x_i, \beta)$
With 16 times more Slight observations, terms corresponding to Slight dominate the sum. The optimization process learns that predicting high probability for Slight (low probabilities for Fatal/Serious) maximizes overall likelihood, even if this means sacrificing all minority class predictions.

Even when the model encounters a true Fatal accident with distinctive features, the learned parameters are so biased toward Slight that predicted probabilities might be P(Fatal)=0.05, P(Serious)=0.10, P(Slight)=0.85, resulting in a Slight prediction. The model has learned patterns, but the imbalance has skewed these patterns irrecoverably toward the majority.

2. LDA: The Prior Probability Problem
LDA explicitly uses class prior probabilities π_k in its discriminant function:
$\delta_k(x) = x^T \Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^T \Sigma^{-1}\mu_k + \log(\pi_k)$

By default, these priors are set to the empirical frequencies from training data: $\pi_Slight = 0.80$, $\pi_Serious = 0.15$, $\pi_Fatal = 0.05$. The $\log(\pi_k)$ term means that Slight receives a boost of $\log(0.80) \approx -0.22$, while Fatal receives $\log(0.05) \approx -2.996$—a difference of 2.78 in the discriminant score.

This means that for an observation to be classified as Fatal rather than Slight, the first two terms of the discriminant function (based on feature values and class means/covariances) must overcome a 2.78-point handicap. In practice, feature-based discrimination is rarely this strong, especially when discriminative features are absent from the dataset.

The equal covariance assumption in LDA also hurts performance. If Fatal accidents genuinely have different variance patterns than Slight accidents (e.g., more extreme values for speed or casualties), LDA cannot capture this, while QDA theoretically could.

3. QDA: Flexibility Without Data
QDA relaxes LDA's equal covariance assumption, allowing each class its own covariance matrix. In principle, this should improve performance if classes truly have different covariance structures. However, QDA requires estimating $p(p+1)/2$ parameters for each class's covariance matrix, where p is the number of features.

With our 11 numeric features (after encoding), this is approximately 66 parameters per class—a total of 198 parameters for covariance matrices alone. For the Fatal class with only 1,252 training examples, this is problematic. The covariance matrix estimates are unstable and potentially singular (non-invertible), leading to numerical issues.

Furthermore, QDA still uses empirical priors just like LDA, so it suffers the same log-probability bias toward the majority class. The added flexibility helps when classes differ substantially in variance structure AND there's sufficient data to estimate parameters reliably. Neither condition is met here: we lack sufficient minority class data, and even if covariances differ, the prior probability effect dominates.

In practice, QDA often performs worse than LDA on small datasets or datasets with class imbalance because the additional parameters increase overfitting risk without sufficient data to estimate them accurately.

4. KNN: The Democracy of Ignorance
KNN is fundamentally a "voting" algorithm—it classifies based on the majority vote among K nearest neighbors. In an imbalanced dataset, this democratic approach becomes deeply problematic.

Consider the feature space representation of our data. Since 80% of all points are Slight, any given region of feature space is likely to contain primarily Slight instances. When a test observation (even a true Fatal accident) is placed in this space, most of its K nearest neighbors will be Slight accidents simply due to prevalence, not because they're truly similar in meaningful ways.

For example, with K=11 (a typical choice), if the test observation's neighborhood reflects the overall 80-15-5 distribution, we'd expect approximately 9 Slight, 2 Serious, and 0-1 Fatal neighbors on average. The majority vote would always favor Slight.

Even if we use a small K (e.g., K=3) to focus on the most local pattern, the density of Slight accidents in most regions means these three neighbors are still likely to be predominantly Slight. KNN would need the feature space to be so well-separated that Fatal and Serious accidents occupy completely distinct regions with no overlap—an unrealistic expectation given the complexity and overlap in real-world accident patterns.

Moreover, KNN doesn't provide a natural mechanism for incorporating class weights or costs, making it particularly unsuitable for imbalanced problems without extensive modifications.
Why All Methods Failed Identically:
The fact that four methods with different theoretical foundations all produced identical predictions (confusion matrices with identical zero entries for Fatal/Serious) reveals that the class imbalance problem is not merely a quirk of one algorithm but a fundamental statistical challenge:

1. **The imbalance ratio (16:1) exceeded the discriminative power of available features**: None of our features were sufficiently predictive to overcome the statistical bias toward the majority class. If we had perfect predictors (e.g., "impact_speed > 120 km/h" always means Fatal), even simple models could succeed. But real-world features are noisy and overlapping.

2. **All methods either explicitly or implicitly used class frequencies**: Whether through prior probabilities (LDA/QDA), likelihood weighting (Logistic Regression), or neighborhood voting (KNN), every method gave disproportionate weight to the majority class.
3. **No method had built-in imbalance correction**: We used standard implementations without class weights, sampling adjustments, or cost-sensitive modifications. This is the default behavior in most statistical software and represents common practice—which is precisely why imbalanced classification is such a pervasive problem.
4. **The optimization objectives mismatched the application requirements**: All methods optimized for overall accuracy or likelihood, which in imbalanced settings means optimizing for majority class performance. None directly optimized for minority class sensitivity, which is what we actually need for traffic safety applications.

Key Lesson:
The uniform failure across diverse methods demonstrates that class imbalance is not a problem that can be solved by trying different algorithms. Rather, it requires fundamental changes to the learning process itself: resampling the data, reweighting the classes, or reformulating the optimization objective. Simply trying alternative standard classification methods (e.g., adding SVM or Naive Bayes to our comparison) would likely yield identical results without these modifications.
This finding underscores the importance of understanding the statistical properties of your data before selecting modeling approaches, and the necessity of specialized techniques for imbalanced problems rather than relying on textbook methods applied without modification.

V. ADDRESSING CLASS IMBALANCE: STRATEGIES FOR IMPROVEMENT

While our implemented models failed to provide useful predictions, this outcome offers valuable lessons about the challenges of imbalanced classification and points toward concrete solutions. This section discusses strategies for addressing class imbalance that could dramatically improve performance in future work.

A. Resampling Techniques

Resampling modifies the training data distribution to reduce or eliminate class imbalance before model training.

1. Synthetic Minority Over-sampling Technique (SMOTE)
SMOTE generates synthetic examples of minority classes by interpolating between existing minority class instances in feature space. For each minority class example, SMOTE:

- Identifies its K nearest minority class neighbors
- Randomly selects one of these neighbors
- Creates a synthetic example at a random point along the line segment connecting the two instances in feature space

This approach increases the representation of minority classes without simply duplicating existing examples, which can lead to overfitting. In R, SMOTE can be implemented using the ROSE or smotefamily packages:

```
library(ROSE)
balanced_data <- ROSE(Accident_Severity ~ ., data = train_data)$data
```

For our dataset, SMOTE could generate synthetic Fatal and Serious accidents to achieve a more balanced distribution, such as 40% Slight, 30% Serious, 30% Fatal. This would give models sufficient examples to learn patterns for all classes.

2. Random Undersampling
Undersampling reduces the size of the majority class by randomly removing instances to balance class proportions. This is computationally simple but risks discarding potentially valuable information. For our dataset with 20,148 Slight accidents in the training set, undersampling might reduce this to ~4,000 to match the minority class sizes more closely.

The main disadvantage is losing 80% of the Slight accident data, which could contain useful patterns. However, for datasets with extreme imbalance and large sample sizes, this may be acceptable.

3. Hybrid Approaches
Combining over- and undersampling often yields the best results. For example, we might apply SMOTE to increase Fatal and Serious accidents to 8,000 examples each, while undersampling Slight accidents to 12,000, achieving a more balanced 30%-30%-40% distribution without the extreme data loss of pure undersampling or the risk of over-synthetic data from pure oversampling.

B. Class Weighting

Class weighting assigns different importance to different classes during model training, effectively penalizing errors on minority classes more heavily.

Implementation Approach:
Weights are typically set inversely proportional to class frequencies:

- weight_Fatal = N_total / (n_classes × N_Fatal)
- weight_Serious = N_total / (n_classes × N_Serious)
- weight_Slight = N_total / (n_classes × N_Slight)

For our data, this yields:

- Fatal weight: 35,999 / (3 × 1,789) ≈ 6.71
- Serious weight: 35,999 / (3 × 5,428) ≈ 2.21
- Slight weight: 35,999 / (3 × 28,782) ≈ 0.42

This means misclassifying a Fatal accident is penalized approximately 16 times more heavily than misclassifying a Slight accident (6.71 / 0.42 ≈ 16).

In R:
For logistic regression using the nnet package:

```
class_weights <- 1 / table(train_data$Accident_Severity)
model <- multinom(Accident_Severity ~ ., data = train_data,
                  weights =
class_weights[train_data$Accident_Severity])
```

Many algorithms including Random Forest, XGBoost, and SVM support class weighting natively, making this one of the most practical approaches for addressing imbalance.

C. Alternative Evaluation Metrics
Moving beyond accuracy to metrics that better capture minority class performance is essential for imbalanced problems.

1. F1-Score
The F1-score is the harmonic mean of precision and recall:
F1 = 2 × (Precision × Recall) / (Precision + Recall)
For our models with 0% recall for Fatal and Serious classes, the F1-score would be 0%, correctly indicating complete failure (unlike accuracy which incorrectly suggests 80% success).

2. Macro-Averaged F1
Macro-F1 computes F1-score for each class separately and averages them, giving equal weight to each class regardless of size:
Macro-F1 = (F1_Fatal + F1_Serious + F1_Slight) / 3
This metric ranges from 0 to 1, where our models would score near 0.27 (0 + 0 + 0.80)/3, more accurately reflecting poor performance than overall accuracy.

3. Class-Specific Sensitivity
For safety-critical applications, we should focus on sensitivity (recall) for minority classes. A model that achieves 70% sensitivity for Fatal accidents and 60% for Serious accidents might be highly valuable even if overall accuracy is only 65%, because it successfully identifies the most critical cases.

4. Receiver Operating Characteristic (ROC) Curves and Area Under Curve (AUC)
ROC curves plot true positive rate (sensitivity) against false positive rate (1-specificity) across all possible classification thresholds. The AUC summarizes overall performance, with values above 0.8 indicating good discrimination. For multiclass problems, we can compute ROC-AUC for each class in a one-vs-rest fashion.
In R:

```
library(caret)
cm <- confusionMatrix(predictions,
test_data$Accident_Severity, mode = "everything")
# This provides precision, recall, F1, and other metrics per
class
```

D. Ensemble Methods
Ensemble methods that combine multiple models often handle imbalanced data more effectively than single classifiers.

1. Random Forest
Random Forest builds multiple decision trees on bootstrapped samples and averages their predictions. It can incorporate class weights and tends to be more robust to imbalance than single decision trees. Balanced Random Forest is a variant that performs bootstrap sampling to ensure balanced class representation in each tree.

2. XGBoost (Extreme Gradient Boosting)
XGBoost is a gradient boosting framework with built-in handling for imbalanced data through the scale_pos_weight parameter, which adjusts the balance of positive and negative weights. It often outperforms traditional methods on imbalanced datasets due to its sequential learning process that can focus on hard-to-classify minority examples.

3. Implementation Considerations
While these advanced methods were beyond the scope of our course requirements (which focused on classical methods from the first four chapters), they represent the state-of-the-art for imbalanced classification and would be the natural next step for this research.

E. Cost-Sensitive Learning
Cost-sensitive learning explicitly incorporates the real-world costs of different misclassifications into the optimization objective. In traffic safety, the cost of failing to predict a Fatal accident (resulting in inadequate emergency response and potentially preventable deaths) is orders of magnitude higher than the cost of incorrectly predicting high severity for a Slight accident (resulting in over-allocation of emergency resources).
A cost matrix might assign:

- Cost of false negative Fatal: 1,000 (extremely high cost—missing a fatal accident)
- Cost of false negative Serious: 100 (high cost—missing serious injuries)
- Cost of false positive Fatal/Serious: 1 (low cost—unnecessary resource dispatch)
- Cost of correct classification: 0

By optimizing for expected cost rather than accuracy, models would prioritize correctly identifying fatal and serious accidents even at the expense of lower overall accuracy.

F. Threshold Optimization
For probabilistic classifiers like logistic regression, the default threshold for classification is typically 0.5 (or maximum probability for multiclass). However, this threshold can be adjusted to favor minority class prediction. For example, if the logistic regression model predicts P(Fatal) = 0.15, P(Serious) = 0.20, and P(Slight) = 0.65 for a given accident, the default rule would classify it as Slight. However, we could adjust thresholds such that any P(Fatal) > 0.10 results in a Fatal prediction, prioritizing sensitivity for this critical class.
Threshold optimization involves systematically varying decision thresholds to maximize a chosen metric (e.g., F1-score for minority classes) on a validation set.

G. Feature Engineering and Additional Data Collection
Improving model performance may also require better features and more data:

1. Additional Features:
    - Driver characteristics (age, experience, license status)
    - Detailed vehicle information (type, age, safety ratings)
    - Road surface conditions and maintenance status
    - Traffic density at time of accident
    - Weather severity (not just categories but specific measurements)

2. Temporal Features:
    - Day of week effects (weekend vs weekday)
    - Seasonal patterns
    - Proximity to holidays or major events

3. Geographic Features:
    - Distance to nearest hospital
    - Urban vs rural classification
    - Historical accident density in the area

4. More Data:
    - Collecting additional accident records, particularly for minority classes
    - Augmenting with similar data from neighboring countries or regions
    - Historical data spanning more years

VI. CONCLUSION

This study investigated the application of four classical classification methods—Logistic Regression, Linear Discriminant Analysis, Quadratic Discriminant Analysis, and K-Nearest Neighbors—to predict traffic accident severity in Bahrain. While all models achieved 79.95% overall accuracy, detailed analysis revealed critical limitations that render them unsuitable for practical deployment.

The primary finding is that severe class imbalance (79.95% Slight, 15.08% Serious, 4.97% Fatal) caused all models to adopt a majority-class prediction strategy, resulting in zero sensitivity for Fatal and Serious accidents. This represents complete failure for the most critical use case: identifying severe accidents requiring immediate emergency response. The Kappa statistic of zero confirmed that models provided no predictive value beyond the baseline of always guessing the majority class.

Key Insights:

1. High Accuracy Can Be Meaningless: Overall accuracy of 79.95% masked complete inability to detect minority classes. This demonstrates that accuracy is an inadequate and potentially misleading metric for imbalanced classification problems.
2. Class Imbalance Requires Specialized Techniques: Standard classification methods, when applied without modification to imbalanced data, consistently fail to detect minority classes. The problem is fundamental: algorithms optimize for overall accuracy, which in imbalanced settings is dominated by the majority class.
3. Minority Classes Are Often Most Important: In traffic safety and many other domains (fraud detection, disease diagnosis, equipment failure prediction), rare events are precisely the cases where prediction would be most valuable. A model that cannot detect these events has no practical utility.
4. Multiple Evaluation Metrics Are Essential: Confusion matrices, class-specific sensitivity, and the Kappa statistic were crucial for understanding model failure. Relying solely on overall accuracy would have led to incorrect conclusions about model effectiveness.

Contributions:

Despite disappointing predictive performance, this study makes several contributions:

- Provides systematic comparison of four classical classification methods on real-world traffic accident data from Bahrain, an understudied region
- Clearly demonstrates challenges of class imbalance and inadequacy of standard methods without appropriate modifications
- Establishes baseline performance and identifies concrete directions for improvement
- Offers educational value by illustrating common pitfalls in applied machine learning

Recommendations for Future Work:

Based on our findings and analysis of class imbalance, we recommend the following approaches for future research:

1. Implement Resampling: Apply SMOTE to oversample minority classes or hybrid sampling to balance the training distribution
2. Use Class Weighting: Assign higher penalties to minority class misclassification (10-20x weight for Fatal accidents)
3. Adopt Ensemble Methods: Implement Random Forest or XGBoost with built-in imbalance handling
4. Change Evaluation Focus: Use F1-score, ROC-AUC, and class-specific sensitivity rather than overall accuracy
5. Apply Cost-Sensitive Learning: Incorporate real-world costs of different misclassifications into the optimization objective
6. Feature Engineering: Develop additional features and collect supplementary data on driver and vehicle characteristics
7. Consider Alternative Problem Formulations: Binary classification (Severe vs Non-Severe) may be more tractable and equally actionable

Final Reflection:

While this study did not produce deployable predictive models, it provides valuable insights into the challenges of real-world classification problems and establishes a foundation for future research. The lessons learned about class imbalance, metric selection, and the gap between statistical accuracy and practical utility are broadly applicable to machine learning projects in public health, safety, and other domains where rare events are of greatest concern.

The path forward requires moving beyond standard textbook methods to specialized techniques designed for imbalanced data. With appropriate modifications—SMOTE, class weighting, ensemble methods, and minority-focused evaluation metrics—we are confident that effective predictive models for traffic accident severity can be developed, ultimately contributing to improved emergency response and reduced casualties on Bahrain's roads.

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